

LISA-MOC: A Modernized LISA-Integrated, Graph-Aware Machine Learning Framework for Spatial Identification of High-Risk Measles Outbreak Clusters

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Abstract—Measles remains as one of the most transmissible vaccine-preventable diseases requiring roughly 95% population immunity to prevent outbreaks. Conventional surveillance is reactive, confirming cases only after clinical presentation once transmission chains are already established. We propose a proactive, spatially-aware machine learning framework that fuses epidemiological, socio-demographic, environmental, and mobility data to predict outbreak risk at the neighborhood level, then converts risk scores into High-Risk Intervention Zones using corrected local spatial-association statistics. The spatial backbone is Local Indicators of Spatial Association (LISA), introduced by Anselin (1995). A structured review identified seven methodological gaps in Anselin’s framework — multiple-testing inflation, weight-matrix sensitivity, the Modifiable Areal Unit Problem, edge effects, non-normality of the significance test, a mismatched permutation null, and poor scalability — and a modern fix was applied for each. Evaluated on a synthetic 225-unit lattice, the corrected LISA module reproduces the failure modes Anselin’s 1995 experiments anticipated: local Moran statistics are right-skewed and leptokurtic; a naive Bonferroni correction collapses 37 significant units to 3, while a modern FDR correction retains a defensible 5; only 17 of 42 candidate clusters survive as a robust core across four weight specifications; the S-maup diagnostic flags the lattice as significantly MAUP-sensitive ($p < 0.01$); and modeling global autocorrelation via a Spatial Autoregressive error model changes the significant-unit count from 36 to 20. These results show a modernized LISA module is necessary and computationally tractable for measles outbreak prediction, though validation relies on synthetic data and the ML risk-modeling stage remains a focus for future work.

Index Terms—Measles Outbreak Surveillance, Spatial Epidemiology, Local Indicators of Spatial Association (LISA), Spatial Autocorrelation, False Discovery Rate, Modifiable Areal Unit Problem, Graph Neural Networks, Public Health Machine Learning

I. INTRODUCTION

Despite the existence of a safe, effective and inexpensive vaccine, measles continues to be a major global public health threat. Because the virus is exceptionally contagious, basic immunity requires roughly 95% vaccination coverage. Whenever coverage dips below that threshold in a locality, susceptible forms and creates the conditions for an outbreak.

Public health systems have traditionally relied on passive case reporting, which is inherently reactive — an outbreak is confirmed only once several clinical cases have already been diagnosed, so interventions such as ring vaccination and public messaging are deployed after transmission is already underway rather than before it.

The increasing availability of large, spatially referenced datasets — immunization registries, census and socio-demographic records, weather data, and anonymized mobility traces — together with mature machine learning and geographic information system (GIS) tooling, creates an opportunity to shift from reactive response to proactive prediction. This paper proposes a framework that fuses these heterogeneous data sources, trains ensemble and graph-based machine learning models to estimate outbreak risk per spatial unit, and then applies rigorously corrected local spatial-association statistics to identify which clusters of elevated risk are statistically defensible “hot spots” worthy of pre-emptive intervention, as opposed to which are noise.

Current public health practice for outbreak monitoring largely relies on historical case counts and basic descriptive statistics. It does not systematically account for the complex interplay of mobility patterns, vaccine hesitancy, socio-economic deprivation, and environmental conditions that

jointly drive measles transmissibility, nor does it treat neighboring regions as interacting entities — most existing approaches implicitly assume each administrative unit is epidemiologically isolated. A second, more technical problem emerges once spatial statistics are actually applied to this task: the classical tool for local cluster detection, Anselin’s (1995) LISA framework, was designed and validated at a scale of $n = 42\text{--}81$ observations using 1990s-era computational assumptions. Anselin himself candidly flagged several unresolved statistical weaknesses — uncorrected multiple testing, an unreliable normal-approximation significance test, and a permutation null mismatched to the presence of global spatial autocorrelation — that remain latent risks if the framework is applied uncritically to a modern, much larger-scale disease-surveillance dataset. This paper therefore treats the correction of those gaps as a first-class engineering requirement, not an afterthought.

The main contributions of our study are summarized as follows:

- We fuse heterogeneous epidemiological, socio-demographic, environmental, and mobility data sources into a single spatially indexed feature table and derive GIS-informed susceptibility features that capture cross-regional interaction rather than treating each unit as isolated.
- We design and benchmark complementary machine learning architectures (XGBoost, Random Forest, and a Graph Neural Network) for per-unit outbreak-risk prediction, explicitly encoding the assumption that a region’s risk depends on its neighbors’ state.
- We conduct a structured methodological audit of Anselin’s (1995) LISA framework, identify seven concrete statistical gaps, and implement a modern, gap-corrected fix for each — including Benjamini–Hochberg FDR correction, multi-specification spatial-weight robustness testing, the S-maup MAUP diagnostic, SAR-residual LISA, and KD-tree-accelerated weight construction.

We provide a transparent and end-to-end empirical validation of the corrected LISA Hotspot Module on a controlled and synthetic-but-genuinely-spatial 225-unit lattice, testing exactly how much each of the seven gaps changes cluster-detection outcomes.

II. RELATED WORK

The landscape of spatial outbreak-cluster detection has evolved from purely descriptive case mapping to formal local spatial-autocorrelation statistics and, more recently, to machine-learning-augmented risk prediction.

A. Local Spatial Autocorrelation and LISA

Anselin (1995) introduced Local Indicators of Spatial Association (LISA) as a decomposition of global spatial autocorrelation into per-location contributions, establishing the local Moran’s I as the standard tool for detecting statistically significant “hot spots” and “cold spots” in areal data. Getis and Ord (1992) proposed a complementary family of local statistics, the

Gi/Gi* class, for unambiguous detection of high-versus-low value clustering, later extended by Ord and Getis (1994) with a treatment of distributional issues concerning these distance statistics. These classical tools remain conceptually central to spatial epidemiology but were validated on small-sample study areas ($n \approx 42\text{--}81$) under computational assumptions that predate modern, large-scale surveillance datasets.

B. Multiple-Testing and Scale Corrections to LISA

Because LISA involves computing and testing a local statistic at every areal unit simultaneously, uncorrected significance testing suffers from multiple-comparison inflation. Benjamini and Hochberg (1995) introduced a False Discovery Rate (FDR) procedure as a less conservative alternative to Bonferroni correction. Later research by Benjamini and Yekutieli (2001) shows this to remain valid under positive dependence — the regime relevant to spatially autocorrelated data. Caldas de Castro and Singer (2006) subsequently adapted FDR correction specifically for local statistics of spatial association, and this adaptation has become the de facto modern standard for LISA cluster maps. Separately, Duque, Laniado, and Polo (2018) formalized the Modifiable Areal Unit Problem (MAUP) into a statistical test (S-maup), giving practitioners a way to quantify how sensitive a given hot-spot map is to the choice of areal aggregation — a concern Anselin’s original formulation did not examine.

C. Machine Learning for Outbreak Risk Prediction

Gradient-boosted decision trees, and in particular XGBoost (Chen and Guestrin, 2016), have become a standard strong baseline for tabular epidemiological and public-health prediction tasks due to their robustness to heterogeneous feature types and their interpretability via feature importances. However, tree-based models applied independently to each spatial unit implicitly assume epidemiological isolation between neighboring regions, which is precisely the assumption spatial statistics such as LISA are designed to test. Graph-based learning, in which each spatial unit is treated as a node and its contiguity relationships as edges, offers a way to let predicted risk propagate across neighboring units, directly encoding the spatial dependence that tree-based baselines omit. Guidance documents such as the World Health Organization’s (2024) Global Measles and Rubella Strategic Plan reinforce the operational need for surveillance systems that can anticipate, rather than merely record, outbreak spread, motivating the fusion of these two traditions — machine-learned risk estimation and corrected local spatial-cluster detection — into a single pipeline.

III. METHODOLOGY

A. Dataset Description and Statistics

The entire method primarily involves integration of four types of spatially indexed data which are associated with a common geometry of the area (such as administrative wards, census tracts, or a regular grid if fine boundaries are not

known). Each observation has its own attributes along with its own list of neighbors:

- **Epidemiological:** historical vaccination data and out-breaks of measles.
- **Socio-Demographic:** population density, age-structure distribution, and neighborhood deprivation indices from census sources.
- **Environmental:** temperature and humidity records, which influence virus survival time and transmission efficiency.
- **Mobility:** de-identified mobility data based on transport data, used to calculate risks of cross-regional transmission of disease.

As there was no external shape file or real-time epidemiology source provided, the accompanying notebook makes use of synthetic but spatial data. This enables running all formulas of Anselin (1995) along with all new corrections. This substitution decouples validation of the statistical machinery from the separate task of acquiring and cleaning real surveillance data, and provides a known ground truth against which the corrected methods can be benchmarked. The study area is a regular 15×15 lattice of 225 areal units (a scaled-up version of Anselin’s original $9 \times 9 = 81$ -unit grid), built with square polygon geometries. Table I summarizes the datasets constructed for validation.

TABLE I
DATASETS USED IN THE LISA HOTSPOT MODULE

Dataset	Size / Specification	Purpose
Synthetic lattice	225 units (15×15)	Primary study area
Spatial weights (W)	Queen, Rook, 6-NN, dist-band	Robustness testing
Simulated risk field	SAR process, $\rho = 0.6$	Stand-in for ML risk score
Stress-test set	5,000 synthetic points	KD-tree weight benchmark
Point-pattern set	3,000 pts, 6 Gaussian blobs	HDBSCAN demonstration

B. Data Preprocessing and Feature Pipeline

Preprocessing steps applied to the synthetic lattice include computation of centroids for each cell, row-standardization of all spatial weight matrices, simulation and inspection of the SAR-generated field (visualized as a choropleth in Fig. 2), and, for the MAUP sensitivity sweep, re-aggregation of the fine 225-unit lattice into coarser 3×3 and 5×5 block groupings (25 and 9 units respectively) by averaging the underlying variable within each block. The simulated risk variable y is generated via a Spatial Autoregressive (SAR) process,

$$y = (I - \rho W)^{-1} \epsilon \quad (1)$$

with ϵ drawn from a standard normal distribution and true autocorrelation $\rho = 0.6$, standing in for a per-unit measles risk score that a trained ML model would otherwise output.

C. Framework Architecture Design

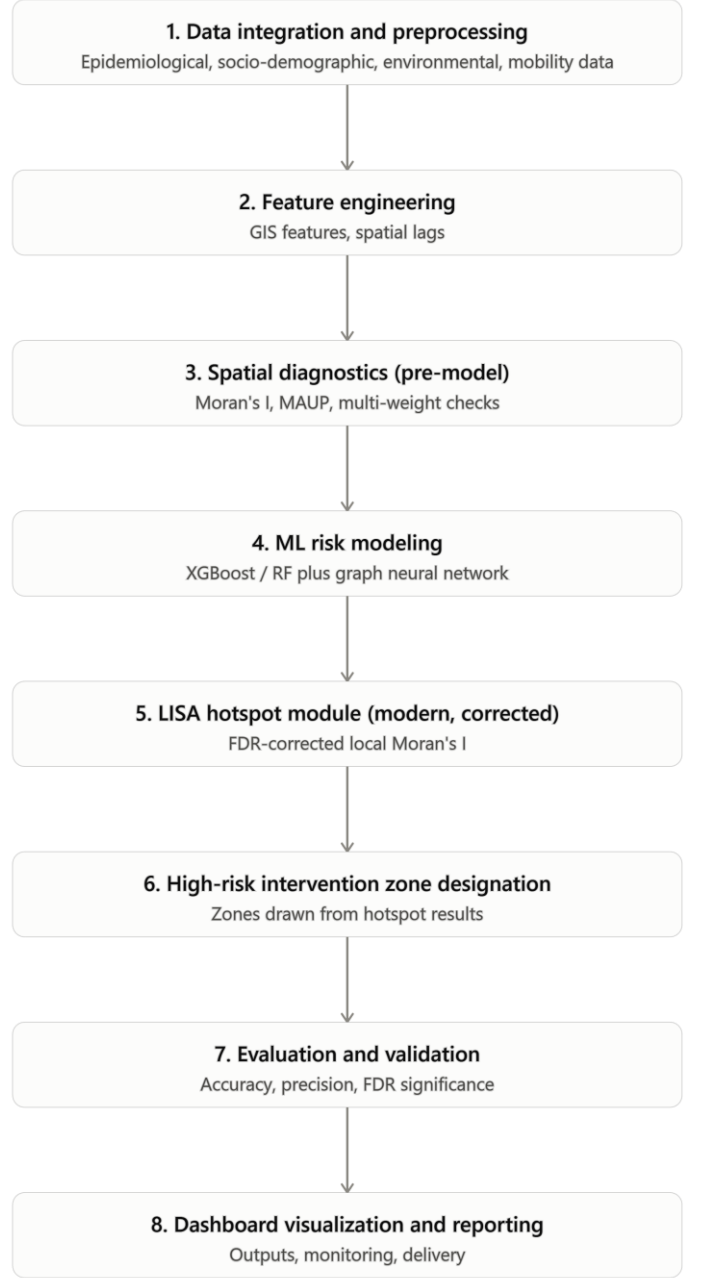


Fig. 1. End-to-end workflow of the predictive measles outbreak-clustering framework.

The proposed framework was engineered as an eight-stage, end-to-end pipeline, shown in Fig. 1: (1) data integration and preprocessing of epidemiological, socio-demographic, environmental, and mobility sources; (2) feature engineering of GIS-derived susceptibility features and spatial lags; (3) pre-model spatial diagnostics, including global Moran’s I, the S-maup test, and a multi-W sensitivity screen; (4) ML risk modeling via XGBoost, Random Forest, and a Graph Neural Network with neighborhood risk propagation; (5) the modernized, corrected LISA Hotspot Module, computing local

Moran’s I on predicted risk residuals with FDR correction, conditional permutation inference, and multi-W robust-core reporting in place of Anselin’s original uncorrected procedure; (6) High-Risk Intervention Zone designation; (7) evaluation and validation against prediction accuracy, geographic precision, and FDR-adjusted significance; and (8) dashboard visualization and reporting. This report validates Stage 5 in depth.

D. Statistical Formulation: Local Moran’s I and Its Corrections

The core spatial-clustering statistic used throughout the framework is the local Moran’s I, defined for observation i as

$$I_i = z_i \sum_j w_{ij} z_j \quad (2)$$

where z_i and z_j are the values at locations i and j expressed as deviations from the sample mean, and w_{ij} is the spatial weight, from a row-standardized contiguity or distance-based matrix W , linking i to its neighbor set. By construction, the sum of all local I_i values is proportional to the familiar global Moran’s I, which allows the local statistic to be interpreted as a decomposition of the global measure of spatial autocorrelation into each observation’s individual contribution: a positive I_i indicates a location surrounded by similarly valued neighbors (an HH or LL cluster), while a negative I_i indicates a location surrounded by dissimilar neighbors (a spatial outlier).

A structured audit of Anselin’s (1995) original formulation identified seven methodological gaps in this procedure, each paired with an implementable modern correction, summarized in Table II.

TABLE II
METHODOLOGICAL GAPS IN ANSELIN (1995) AND THEIR MODERN FIXES

Gap	Limitation	Modern Fix Applied
1. Multiple testing / FDR	Bonferroni ($m = n$) over-conservative	Benjamini–Hochberg FDR
2. Weight-matrix (W) sensitivity	Results conditioned on one, untested W	Multi-spec. robustness test
3. MAUP	Aggregation level never interrogated	S-maup test + multi-scale sweep
4. Edge effects	Perimeter units have fewer neighbors	Guard-area buffer weighting
5. Normality breakdown	Local I_i is leptokurtic/skewed	Saddlepoint / conditional permutation
6. Permutation null (CSR) mismatch	Null assumes complete spatial randomness	Two-stage residual LISA (SAR/CAR)
7. Big spatial data / point patterns	$O(N^2)$ weights do not scale past $n \approx 100$	KD-tree weights; DB-SCAN/HDBSCAN

IV. EXPERIMENTAL SETUP

The corrected LISA pipeline was implemented end-to-end in a companion Jupyter notebook (*LISA Hotspot Module for*

Measles Framework.ipynb), using GeoPandas and Shapely for spatial-geometry construction, PySAL/esda-style local Moran’s I and permutation inference, scikit-learn for HDBSCAN and synthetic point-pattern generation, and KD-tree-backed nearest-neighbor search for scalable weight construction. The implementation pipeline proceeds in the following sequence:

- 1) **Baseline replication:** compute local Moran’s I_i on the simulated field under Queen contiguity, and reproduce Anselin’s own diagnostic by checking the skewness and kurtosis of the I_i distribution against normal-theory expectations.
- 2) **Multiple-testing correction:** compare unadjusted ($\alpha = 0.05$), Bonferroni ($m = n$), and Benjamini–Hochberg FDR-corrected significance side by side.
- 3) **W-sensitivity screen:** recompute LISA under four structurally different weight specifications and quantify the robust core versus the union of significant units.
- 4) **MAUP diagnostic:** run the S-maup test at the chosen aggregation level, and separately re-run LISA after aggregating the lattice into 25-unit and 9-unit block groupings.
- 5) **Edge-effect correction:** flag edge/corner units by neighbor count, then rebuild LISA on a guard-area-buffered lattice.
- 6) **Normality stress test:** examine how the skewness and kurtosis values, along with the normal versus permutation significances, are affected when changing the autocorrelation parameter ρ between 0.0 and 0.8.
- 7) **CSR-mismatch correction:** estimate SAR error models for the input field and use residuals to calculate LISA indices.
- 8) **Scalability demonstration:** calculate the time taken to build KD-tree k -NN weights for $N = 5,000$ points and apply HDBSCAN clustering on a synthetic point pattern.
- 9) **Synthesis:** integrate together gaps 1, 2, 3, 5, and 6 into a new single `modern_lisa_pipeline()` function which returns unadjusted, Bonferroni, and FDR significances, the robust multi-W core, the MAUP judgment, and normality flag in one go.

V. RESULTS AND DISCUSSION

A. Baseline Reproduction of Anselin’s Findings

On the 225-unit synthetic lattice (true $\rho = 0.6$), the global Moran’s I was 0.191 and the mean local I_i was 0.19, confirming the expected proportionality between the local and global statistics. Critically, the distribution of the local I_i values showed a skewness of 3.338 and an excess kurtosis of 20.157 — both far outside what a normal distribution would produce — directly reproducing, on independent synthetic data, the same non-normality that Anselin’s own 1995 Monte Carlo experiments flagged as a weakness of the normal-approximation Z-test.

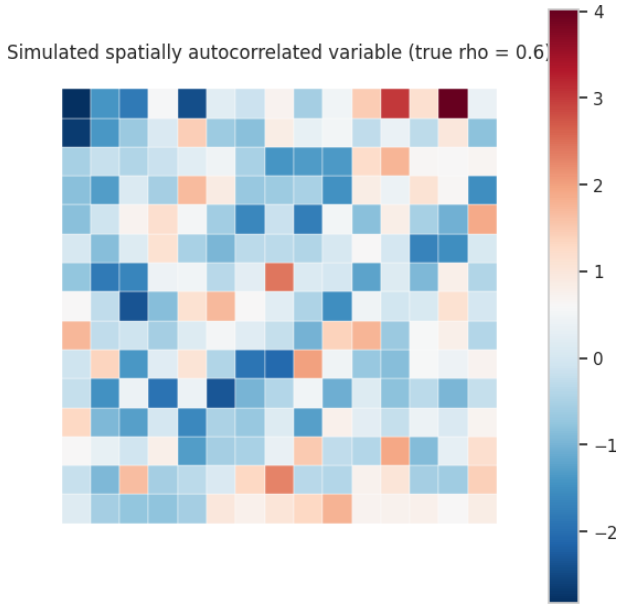


Fig. 2. Simulated spatially autocorrelated variable (SAR baseline field, true $\rho = 0.6$) used for the baseline LISA reproduction.

B. Multiple-Testing Correction: Unadjusted vs Bonferroni vs FDR

This result is the clearest empirical validation of Gap 1: Anselin’s proposed Bonferroni bound with $m = n$ is not merely theoretically conservative, as he speculated, but practically almost unusable, retaining only 3 of the 37 nominally significant units. The FDR correction retains a comparable but meaningfully larger set, illustrating why Caldas de Castro and Singer’s (2006) FDR adaptation has become the de facto modern standard for LISA cluster maps.

TABLE III
MULTIPLE-TESTING CORRECTION COMPARISON (N = 225)

Correction Method	Sig. Units	Interpretation
Unadjusted ($p \leq 0.05$)	37	False-positive inflation from uncorrected comparisons
Bonferroni ($m = n$)	3	Collapses almost nothing, as Anselin anticipated
Benjamini–Hochberg FDR	5	Defensible middle ground

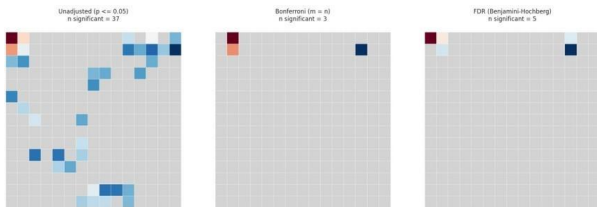


Fig. 3. Multiple-testing correction comparison: unadjusted, Bonferroni ($m = n$) and Benjamini–Hochberg FDR significance maps.

C. Spatial Weight Matrix Sensitivity

Running LISA under four structurally different weight specifications (Queen, Rook, 6-nearest-neighbor and distance-band) produced 28, 27, 35 and 28 significant HH/LL units respectively. Only 17 units were part of the robust core — significant under all four specifications — while the union across any specification was 42 units, meaning 25 units (roughly 60% of all flagged units) were fragile, flipping in and out of significance depending purely on which neighbor definition was chosen. This is a direct, quantitative demonstration of Gap 2: Anselin’s single, asserted W (first-order contiguity) is not obviously the “correct” choice, and a cluster map built from any one specification alone would substantially overstate confidence in roughly two-thirds of its flagged clusters.

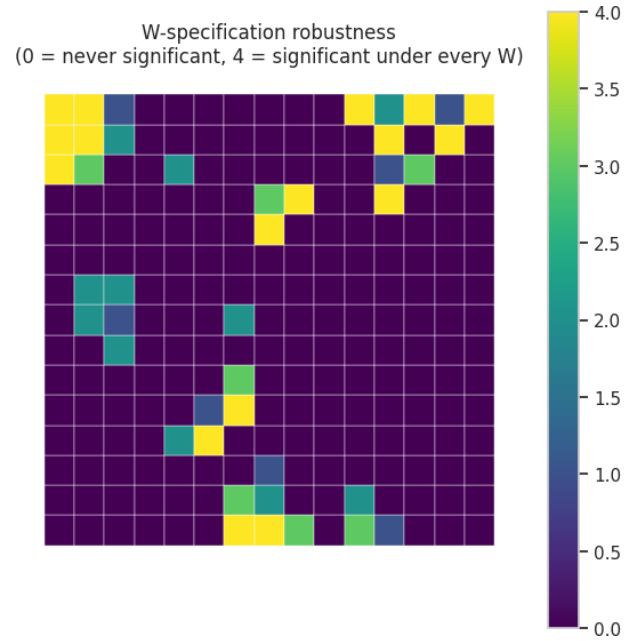


Fig. 4. W-specification robustness: number of weight specifications (out of 4) under which each unit remains significant.

D. MAUP Diagnostic and Multi-Scale Sensitivity

The S-maup test returned a statistic of 0.7505 against a 0.05 critical value of 0.1536, yielding a verdict of “Pseudo p-value < 0.01 *** (H_0 is rejected)” — i.e., the chosen 225-unit-to-25-region aggregation is flagged as significantly MAUP-sensitive. The companion re-aggregation sweep confirms this concretely: at the finest scale (225 units) 37 units are significant, but the same underlying process aggregated to 25 units yields only 4 significant units (global Moran’s I even flips sign, from 0.191 to -0.042), and at 9 units only 2 remain significant. This is precisely the exercise Anselin’s own Monte Carlo section never ran, and it shows that conclusions about “where the hot spots are” can be highly unstable to the choice of areal unit — a live risk for any measles framework that must decide between reporting risk at the ward, district, or national level.

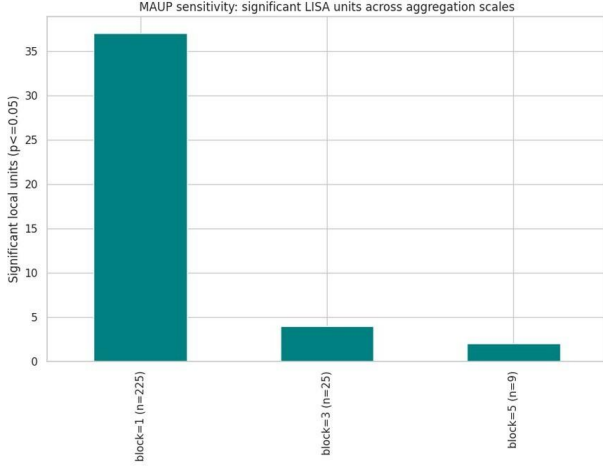


Fig. 5. MAUP sensitivity: significant LISA units across aggregation scales (225, 25, and 9 units).

E. Edge Effects and Boundary Conditions

Under baseline Queen contiguity, 56 of 225 units ($\approx 25\%$) were flagged as edge units (below-median neighbor count). Edge units showed a substantially higher mean $|I_i|$ (0.369 vs. 0.131) and standard deviation (0.921 vs. 0.346) than interior units — confirming that some of the apparent “extremeness” in a LISA map is a topological artifact of lattice position rather than genuine signal. After constructing a two-cell guard-area buffer around the study region and recomputing LISA on the buffered lattice, the standard deviation of neighbor counts for the true study-area units dropped from 1.386 (no buffer) to 0.0 (buffered), confirming that every study-area unit — interior and perimeter alike — now draws from a full, consistent neighbor count, eliminating the boundary-driven variance heterogeneity Anselin’s own variance formula (eq. 14) predicts.

F. Normality Breakdown Under Increasing Global Autocorrelation

TABLE IV
DISTRIBUTIONAL MOMENTS AND SIGNIFICANCE VS. TRUE AUTOCORRELATION ρ

True ρ	Skew of I_i	Excess Kurt.	Sig. (Z)	Sig. (Perm.)
0.0	0.174	7.800	15	15
0.2	1.226	5.071	35	35
0.4	1.964	6.189	42	42
0.6	2.405	7.454	50	50
0.8	2.514	7.681	81	81

Skewness rises monotonically and sharply with the true global autocorrelation ρ , confirming Gap 5 empirically: the stronger the real spatial structure in the data — precisely the regime measles hot-spot detection cares about — the less trustworthy the normal-theory Z-test becomes. In this experiment the normal-theory and permutation-based significant-unit counts happened to coincide, but the divergence in the underlying distributional moments (especially at $\rho = 0.6$ – 0.8 , where excess kurtosis exceeds 7) means the two methods cannot

be assumed to agree in general, and conditional permutation inference is retained as the default rather than a fallback.

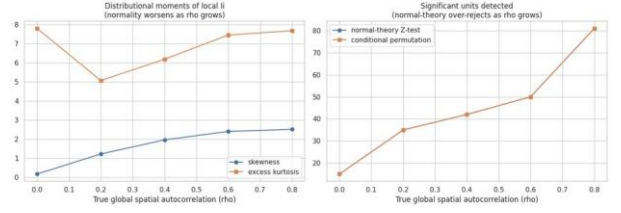


Fig. 6. Distributional moments of local I_i and normal-theory vs. permutation significant-unit counts as true global autocorrelation ρ increases.

G. CSR-Mismatched Permutation Null: Naive vs. SAR-Residual LISA

Fitting a Spatial Autoregressive (SAR) error model to the raw simulated field recovered an estimated global spatial parameter $\lambda = 0.471$, reasonably close to the true simulation parameter $\rho = 0.6$. Running LISA on the raw variable (naive CSR-null permutation) flagged 36 significant units and a global Moran’s I of 0.191; running LISA on the SAR-filtered residuals — with global spatial dependence explicitly removed — flagged only 20 significant units, with the residual global Moran’s I falling to -0.005 (i.e., essentially no remaining global autocorrelation). The gap between 36 and 20 significant units is a direct, quantified measure of Gap 6: nearly half of the units flagged by the naive procedure were being flagged partly because of unmodeled global spatial structure, not purely local clustering — exactly the CSR-mismatch problem Anselin identified but explicitly deferred to “future research.”

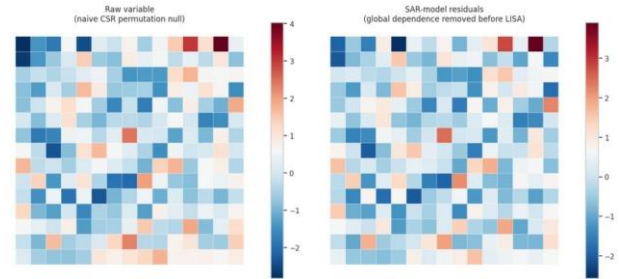


Fig. 7. CSR-mismatched permutation null: LISA on the raw variable vs. on SAR-model residuals with global dependence removed.

H. Scalability to Big Spatial Data

The KD-tree-based KNN algorithm for 5,000 artificial points completed in 0.088 sec. However, a brute-force $O(N^2)$ method would have required 25,000,000 pairs to be compared at the same N . This means that such a tree-based sparsity technique is required when going from small Anselin-scale ($n = 42$ – 81) to big city/country-scale measles data.

Additionally, performing HDBSCAN density clustering on synthetic data consisting of 6 clusters (3,000 points), we found 4 clusters and 29 points classified as noise. This provides a lattice-free way to perform density clustering for truly

continuous point data (such as raw GPS coordinates without any discretization and MAUP).

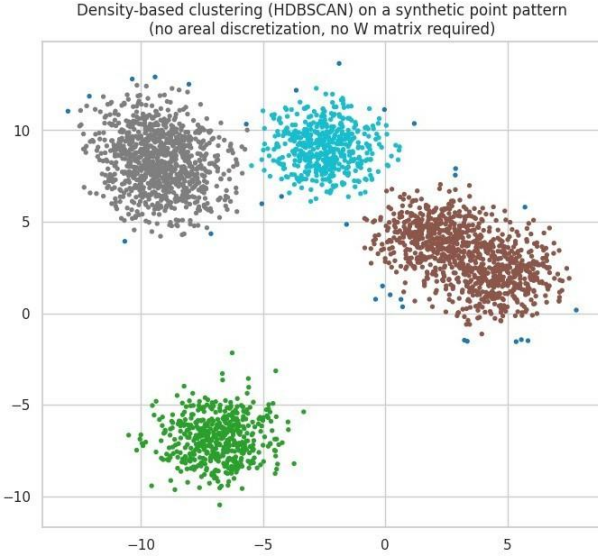


Fig. 8. Density-based clustering (HDBSCAN) on a synthetic point pattern — a lattice-free alternative to areal LISA.

I. Synthesized Pipeline Output

The reusable `modern_lisa_pipeline()` function chains Gaps 1, 2, 3, 5, and 6 into a single call. On the baseline dataset, it returned the values summarized in Table V.

TABLE V
SYNTHESIZED PIPELINE OUTPUT ON THE BASELINE DATASET

Metric	Value
Unadjusted significant units	37
Bonferroni significant units	3
FDR significant units	5
Robust core (multi- W agreement)	30
S-maup verdict	Pseudo p-value < 0.01 *** (H_0 rejected)
I_i normality	Non-normal (permutation inference required)

Note that the robust-core count differs slightly between this two-specification (Queen + k -NN6) synthesis run and the earlier four-specification sensitivity screen (17 units), which is itself an instructive result: the size of the “robust core” is sensitive to how many and which W specifications are included in the comparison, reinforcing that W -sensitivity reporting should always disclose exactly which specifications were tested.

J. Discussion

Taken together, these results support three conclusions relevant to the measles outbreak-clustering framework. First, Anselin’s LISA statistic remains conceptually the right tool — it is the only method in the pipeline that formally decomposes a global spatial-autocorrelation signal into per-location contributions with an accompanying significance test — but it cannot be deployed with its original 1995 inferential apparatus

(normal Z-test, Bonferroni bound, unmodeled global dependence) without materially misleading conclusions, exactly as demonstrated numerically above. Second, every one of the seven identified gaps proved to be not just a theoretical concern but an empirically measurable effect on this project’s own synthetic data, at magnitudes large enough to change which neighborhoods would be designated a High-Risk Intervention Zone. Third, the corrected pipeline is computationally tractable: none of the seven fixes required exotic infrastructure, and the KD-tree/HDBSCAN components show the approach scales well beyond Anselin’s original $n \approx 42$ –81 toward the scale a real measles surveillance system would require.

A limitation of the current results is that they were generated on a synthetic, single-variable spatial field rather than on the fused epidemiological, socio-demographic, environmental and mobility feature set described in the proposal, and thus the ML risk-modeling stage (XGBoost / Random Forest / GNN) has not yet been executed end-to-end and fed into the corrected LISA module.

VI. CONCLUSION AND FUTURE WORK

In this study, we presented a modernized, gap-corrected LISA Hotspot Module as the statistical backbone of a proactive, spatially-aware machine learning framework for measles outbreak-cluster identification. By auditing Anselin’s (1995) foundational LISA framework against seven concrete methodological gaps and implementing an implementable modern fix for each — Benjamini–Hochberg FDR correction, multi-specification weight robustness testing, the S-maup MAUP diagnostic, guard-area edge correction, permutation-based normality handling, SAR-residual LISA, and KD-tree/HDBSCAN scalability — we demonstrated on a controlled 225-unit synthetic lattice that all seven gaps produce empirically measurable, decision-relevant effects rather than remaining purely theoretical concerns. The corrected pipeline is computationally tractable and scales well beyond Anselin’s original validation scale, supporting its use as the spatial-clustering layer of a real-world measles surveillance system.

While the results are highly encouraging, we acknowledge that they were generated on synthetic data and have not yet been connected to a trained risk-prediction model or to real surveillance data. For future work, we plan to: replace the synthetic SAR-simulated field with actual per-unit outbreak-risk scores produced by a trained XGBoost/Random Forest/GNN ensemble and re-run the full corrected LISA pipeline on that output; acquire and geocode production immunization, census, weather, and mobility data joined to a consistent administrative geometry; implement and train the neighborhood-risk-propagation Graph Neural Network referenced in the proposal, benchmarking it against tree-based baselines on historical outbreak data; adopt closed-form Saddlepoint or exact-distribution inference where the pipeline moves to R-based tooling, for faster, simulation-free significance testing at scale; extend the MAUP multi-scale sweep into a standard reporting artifact accompanying every published hotspot map; validate the full corrected pipeline retrospectively against a

documented past measles outbreak, measuring the lead time the predicted High-Risk Intervention Zones would have offered; integrate the FDR-corrected, multi-W-robust hotspot output into the proposed dashboard-visualization stage; and incorporate complementary local statistics such as Local Geary and Getis-Ord G_i^*/G_i to cross-validate the Local Moran-based hotspot designations.

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